

Real-Time Driver Drowsiness Detection System using Vision Transformer for Accurate Eye State Analysis

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Abstract --- Driver fatigue and drowsiness pose significant risks to road safety, necessitating advanced monitoring systems to mitigate accidents. This research addresses this concern through the implementation of a Real-Time Driver Drowsiness Monitoring system using Vision Transformer (ViT). The study involved fine-tuning a Vision Transformer (ViT) model with additional layers on a dataset comprising 84900 images of open and closed eyes, enabling accurate detection of driver drowsiness. The motivation for drowsiness detection stems from the critical role of attentiveness in driving safety. Existing alarm systems often rely on conventional methods, and the research seeks to enhance this by incorporating state-of-the-art ViT technology. Achieving an impressive accuracy of 98.8%, the real-time monitoring system uses the ViT model to detect driver drowsiness promptly. When drowsiness is detected, an alarm is triggered to alert the driver, promoting immediate attention, and reducing the risk of accidents. The model's performance is also thoroughly evaluated, demonstrating high precision, recall, and F1-Score for both closed and open-eye states. This study benefits the field of driver safety by providing a reliable and effective real-time drowsiness monitoring method, thus enhancing road safety, and reducing the incidence of fatigue-related accidents.

Keywords- Vision Transformer (ViT), Driver fatigue, Drowsiness detection, Fine-Tuning, Computer vision, Image processing.

I. INTRODUCTION

Ensuring everyone's safety on the road depends critically on the alertness of drivers. According to estimates from the World Health Organization, 1.3 million people die in accidents each year. Twenty to fifty million more people suffer non-fatal injuries, many of which result in lifelong disability. The main reasons for traffic accidents are tiredness, drowsiness, and inattentive driving. As a result, a system that can alert drivers to fatigue and help them become more vigilant by sending out timely warnings may be able to reduce the number of accidents that occur, save money, and reduce personal suffering.

The research introduces an innovative approach to advance driver safety through the fine-tuning of the Vision Transformer (ViT) model on a dataset for the accurate analysis of driver drowsiness. By delving into advanced computer vision, the study seeks to develop a robust system for proactive detection of driver drowsiness, ultimately contributing to accident prevention and overall road safety improvement. It is worth

emphasizing the preference for Vision Transformer (ViT) over traditional algorithms like CNN or KNN due to ViT's exceptional ability to capture intricate long-range dependencies in images. To ensure reliable training and testing, the model is trained using a large, labeled dataset approach, with the data divided into 80% for training, 10% for validation, and 10% for testing.

II. LITERATURE SURVEY

The authors address driver drowsiness detection using deep learning models with a focus on real-time applications using the ZJU dataset and achieved a 95% accuracy on the task [1]. The primary algorithms employed are modified versions of VGG-19 and VGG-16. The study describes a computer vision-based real-time drowsiness detection system for drivers [2]. It detects drowsiness by measuring eye blink rate and yawning. By detecting eye blink rate, the system obtains 90% accuracy, and by detecting yawns, 60% accuracy. The suggested system uses the Haar Cascade Classifiers to detect faces and eyes, and it sets an alarm level based on the rate at which an eye blinks and yawns.

An enhanced driver drowsiness detection model that makes use of eye image characteristics like circularity and black ratio is presented in the paper [3]. Four models of Support Vector Machines were created, with an accuracy of 91.3%. The paper presents a real-time driver drowsiness detection system that uses visual behavior analysis and machine learning [4]. It achieves a sensitivity of 95.58% and specificity of 100% using Support Vector Machine. The algorithms employed include Support Vector Machine (SVM), Fisher's linear discriminant analysis (FLDA), and the Bayesian classifier. The paper presents a real-time driver drowsiness detection system using gradient statistics and machine learning [5]. It accurately detects eye closure, achieving an accuracy of 91.49% with glasses and 95% without glasses, outperforming a prior method.

Facial landmarks are analyzed using Dlib's form detector, and drowsiness is tracked through studying aspect ratios of the mouth and eyes [6]. The system achieves its highest accuracy on a reference dataset, which is 96.71%. For eye and mouth detection, the system makes use of Dlib's facial landmark detection and aspect ratio computations. The purpose of this study was to assess the effectiveness of two algorithms for

drowsiness detection utilizing a single EEG channel: Artificial Neural Networks (ANN) and Support Vector Machines [7]. Characteristics are extracted using the Fast Fourier Transform (FFT) using the MIT-BIH Polysomnographic dataset. Based on a single EEG channel, it is found that the ANN classifier performs better for EEG drowsiness detection than the SVM classifier. The research introduces an eyes status detection system that makes use of a lightweight convolutional neural network [8]. Compared to other well-known classification networks, the eyes network uses a significantly smaller number of parameters to obtain an accuracy of 98.70%. The paper presents a real-time computer vision system for detecting human drowsiness [9]. Impressive accuracy is achieved by combining drowsiness detection, eye classification techniques, and infrared camera images in the algorithm.

The paper presents a driver drowsiness detection system using the PERCLOS method [10]. It utilizes the Viola-Jones detector, assessing eye closures to detect drowsiness. This research developed a system for detecting driver drowsiness through image processing [11]. The Eye Aspect Ratio (EAR) was utilized to track tiredness and sound an alert. The system achieved an accuracy of approximately 80% and employed facial landmark detection. This research examines deep learning with convolutional neural networks (CNN) for the identification of driver drowsiness [12]. It highlights advantages and disadvantages of classification techniques like SVM and HMM as well as obstacles. Paper presents a driver drowsiness detection model using facial landmarks [13]. The system achieved up to 92.5% accuracy in detecting drowsy drivers. This research presents a method for driver drowsiness detection using a hand-crafted face image descriptor and a supervised classification algorithm [14]. The study uses the NTH Drowsy Driver Detection dataset and reports an accuracy of 75.90%.

III. METHODOLOGY

A. Data Collection and Dataset.

The dataset was curated to encompass images reflecting the real-world complexity of driver drowsiness, including variations in lighting conditions such as shady and dark environments. This section discusses the techniques employed to mitigate the challenge of class imbalance leading to biased model predictions favoring majority class.

1) Data Overview and Organization.

The driver drowsiness detection dataset is large, consisting of approximately 84900 images. The initial steps were to organize the dataset in a way that would make analysis and model training easier. This involved extracting file names and their corresponding labels from the file paths, providing a systematic structure for the data.

2) Class Imbalance and Oversampling of Minority Class.

Upon meticulous analysis of the dataset, significant imbalances were identified in the distribution of classes, particularly concerning open and closed-eye images. Specifically, random oversampling was implemented to augment instances of the minority class, represented by closed-eye images. This meticulous oversampling technique was very helpful in creating a more evenly distributed set of closed-eye

and open-eye images. The dataset gave both classes a more equitable and representative set of examples by deliberately increasing the number of closed-eye cases. As a result, during the subsequent training phase, the model was exposed to more varied instances, which improved performance and generalization.

3) Data Augmentation.

To enhance the versatility of the model and prepare it for various real-world situations, the training images are adjusted. The training images are resized to a standard size of 224 pixels, aligning with the requirements of the Vision Transformer (ViT) model. This ensures that all images processed by the model have the same format. Random rotations up to 30 degrees are used to represent content variations and introduce spatial diversity. The integration of RandomResizedCrop function adds a spatial dimension by randomly cropping and resizing images, allowing the model to focus on different regions during training. For RandomResizedCrop, the scale values are set to (0.8, 1.0), signifying that the cropped area's size can vary between 80% and 100% of the original image. The augmentation pipeline also adds RandomAdjustSharpness function, which introduces random adjustments to the sharpness of images. The factor for RandomAdjustSharpness is set to 2, which denotes the highest sharpness adjustment factor. These augmentations, in conjunction with others, result in a comprehensive augmentation approach that cultivates a more enriched training dataset.

B. Label Encoding.

In the process of label encoding for the open and closed eye image classification task, the identification and categorization of labels were initiated as shown in Figure 1, representing states such as "closedEye" and "openEye". These labels, sourced from the dataset, denote distinct visual conditions captured in the images. Following this, the compilation of these unique labels into a comprehensive list formed the basis for creating two dictionaries: label2id and id2label.

CLOSED EYE	0	0	CLOSED EYE
OPEN EYE	1	1	OPEN EYE

Label to ID Mapping

ID to Label Mapping

Fig. 1. Label to ID and ID to Label Mapping

Every distinct label was given a numeric ID through an iterative mapping process, establishing a bidirectional mapping system to the model's understanding of categorical data. This method creates a smooth link between string representations of eye states and their numerical indices, while also improving readability during training and prediction. Accurate classification of various eye conditions is ensured by matching the model's output labels with the distinct categories in the dataset.

C. Model Selection and Customizing Architecture.

Unlike conventional neural networks (CNNs), the Vision Transformer (ViT) is a practical option in the application of real-time drowsiness detection. ViT can capture complex long-range dependencies within images by processing images as a

sequence of patches and utilizing self-attention mechanisms. Because of its distinctive architecture, ViT is well-suited to recognize subtle patterns in facial expressions that indicate drowsiness, which is important for making quick and precise predictions in real-time scenarios.

The 'google/vit-base-patch16-224-in21k' ViT model was specifically chosen because it has multiple layers and a transformer-based architecture, which is excellent at identifying global dependencies in images because it takes the whole context into account rather than just concentrating on local features. This comprehensive knowledge is useful for detecting drowsiness in a variety of settings and facial expressions. Furthermore, the model's ability to scale to various image resolutions makes it possible to handle image quality variations effectively, which is an essential feature for real-world applications.

1) Model Extension with fully connected layers.

An extension of the architecture is made, as seen in Figure 2, where the ViT model is displayed for the task of drowsiness detection. Additional fully connected layers are added to the original classifier to give the model more depth and ability to learn complex hierarchical representations.

The need to introduce non-linearity and prevent overfitting during the training process led to the selection of fully connected layers with 128, 256, and 384 hidden units for layers 1, 2, and 3 respectively, ReLU activation functions, and a dropout rate of 0.5. The incorporation of these elements seeks to strike a balance between model complexity and generalization performance.

2) Class Label Mapping for Inference.

The setup of the `model.config.label2id` and `model.config.id2label` is crucial to the architecture of the extended Vision Transformer (ViT) model and goes beyond simple indexing in the inference stage. The drowsiness detection system's interpretability and user-friendliness are enhanced by this configuration attribute.

The `model.config.label2id` attribute essentially acts as a bi-directional mapping, associating each class label with a unique numerical identifier. This mapping is essential for both aligning the model's output with the predefined class labels and interpreting the predictions of the model. Similarly, a reverse mapping is offered by `model.config.id2label`, which converts numerical indices back into the appropriate class labels. This bidirectional mapping ensures that the model's predictions are not only numerically coherent but also readily interpretable without manual intervention. This interpretability is especially vital in real-time applications, where quick decision-making is important.

D. Model Training.

The customized ViT model is fine-tuned on a dataset of 67,920 training images. For training under the dataset, the critical hyperparameter, the learning rate, is set at 0.00001, influencing the model's convergence during optimization. To balance computational efficiency and model generalization, batch sizes have been set to 64 for training to effectively leverage parallel processing during the training phase. For validation and testing, a batch size of 32 has been chosen. A

weight decay of 0.02 is used to limit model parameters during training for better generalization and to prevent overfitting. The learning rate scheduler incorporates a warm-up strategy with 50 warm-up steps, contributing to stability and convergence. To ensure that the model with the best performance is used, model checkpoints are saved at the end of each epoch, and the model with the greatest efficiency is loaded at the end of training.

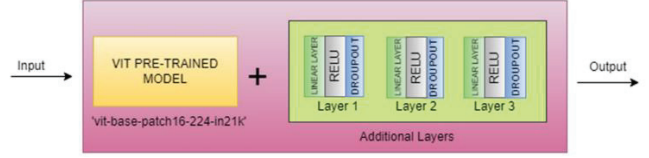


Fig. 2. Vision Transformer Model with Additional Layers

IV. REAL-TIME RECOGNITION

The model used by the real-time eye detection system has been optimized using a proprietary dataset. ViT model with additional layers is trained to identify unique eye states, like 'openEye' and 'closedEye.' The real-time video input is taken from a webcam and provide immediate feedback on the person's eye condition. The trained model parameters and class mappings are stored in the model directory, from which the ViT model and its configuration are loaded. This tuned model demonstrates improved ability to recognize subtle eye states, which improves the accuracy of the system in practical uses. The system connects to the webcam and continuously records video frames using OpenCV. Faces in every frame are recognized using a Haarcascades face detector, which serves as the foundation for further analysis. Here Figure 3 shows the experimental results for detecting the closed eyes or drowsiness of a driver in a car.

For each detected face, the system performs feature extraction using the model feature extractor. These features are then input into the model for prediction. The predicted class index is interpreted, and the corresponding label indicates whether the eyes are predicted to be open or closed. The confidence level for drowsiness predictions is determined by setting a threshold for closed eyes probability at 0.5. The video feed dynamically displays the predictions and real-time results, along with visual indicators that demonstrate whether the eyes are open or closed. The user can quickly and easily understand the system's analysis of this visual feedback. In addition, for enhanced safety, if the system detects drowsiness an alarming sound is triggered to quickly capture the driver's attention and prevent potential lapses in alertness.

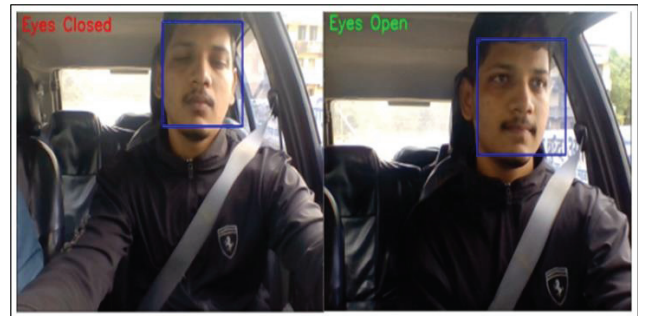


Fig. 3. Real-Time Detection of driver in a car for detecting closed eyes

V. DISCUSSION

The model underwent training for 10 epochs, employing a small batch size of 64 during the training phase. The dataset was partitioned into sets: 80% for training, 10%, for validation, and 10% for testing. Throughout the training process, the model exhibited robust learning dynamics. Upon evaluation of the test data, the model demonstrated an exceptional accuracy of 98.8%. This accuracy, attained with a validation and testing batch size of 32, confirms the model's ability to produce reliable and accurate predictions by highlighting its effectiveness in identifying and utilizing crucial patterns within the dataset. The model's high accuracy not only demonstrates how well it learned complex features from the training data, but also shows that it has the potential to make accurate and dependable predictions in real-world situations. Below Table 1 shows the decreasing training and validation loss over increasing epochs.

TABLE I. DECREASING TRAINING AND VALIDATION LOSS

Epoch	Training Loss	Validation Loss
1	0.084800	0.079997
2	0.058500	0.048371
3	0.038600	0.037518
4	0.032800	0.032195
5	0.025200	0.026430
6	0.021500	0.024348
7	0.017400	0.021178
8	0.017200	0.019996
9	0.017900	0.019868
10	0.016500	0.019380

Here, Table 2 summarizes the model's performance in closed-eye and open-eye detection, with a focus on Precision, Recall, and F1 Score for each detected state.

TABLE II. EYE STATE CLASSIFICATION MATRIX

Eye State	Precision	Recall	F1-Score
Closed eye	99.51%	98.97%	99.15%
Open eye	99.91%	98.81%	99.16%

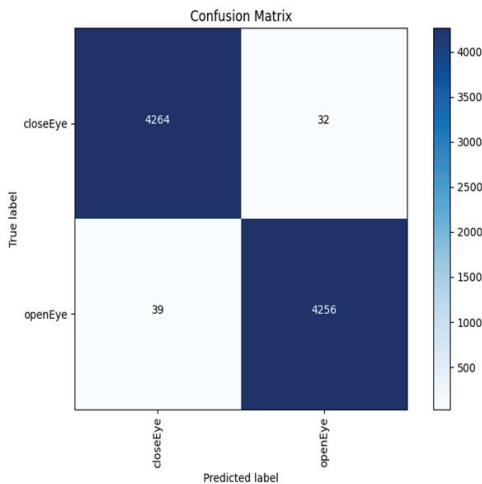


Fig. 4. Confusion Matrix for Model Classification Performance

To provide a deeper insight into the model's performance, a confusion matrix was constructed. The confusion matrix allows assessment of the model's ability to correctly classify samples into their respective classes. Figure 4 provides a summary of the findings.

VI. CONCLUSION

The research has successfully developed a real-time Driver Drowsiness Monitoring system for drivers, employing a combination of image processing techniques and a Vision Transformer (ViT) with additional layers to discern the drowsiness of drivers. With an impressive accuracy rate of 98.8%, the model, trained on an 80-10-10 data split, showcases its robustness in identifying drowsiness. Furthermore, the integration of an alarming sound feature enhances the system's practicality by promptly capturing the driver's attention. With its proactive monitoring of driver attentiveness, this innovative solution represents a major advancement in road safety.

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